

```

import tensorflow as tf
from tensorflow.keras import datasets, layers, models

# Load dataset
(train_images, train_labels), (test_images, test_labels) =
datasets.cifar10.load_data()

# Normalize pixel values
train_images, test_images = train_images / 255.0, test_images / 255.0

# Build a simple CNN model
model = models.Sequential([
    layers.Conv2D(32, (3,3), activation='relu', input_shape=(32,32,3)),
    layers.MaxPooling2D((2,2)),
    layers.Conv2D(64, (3,3), activation='relu'),
    layers.MaxPooling2D((2,2)),
    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])

# Compile the model
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# Train the model
model.fit(train_images, train_labels, epochs=5,
          validation_data=(test_images, test_labels))

# Evaluate the model
test_loss, test_acc = model.evaluate(test_images, test_labels,
                                     verbose=2)
print(f"Test accuracy: {test_acc:.2f}")

```

Downloading data from <https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz>
170498071/170498071 ————— **4s** 0us/step

/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113:
UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using
Sequential models, prefer using an `Input(shape)` object as the first layer in the model
instead.

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

Epoch 1/5

1563/1563 ————— **73s** 45ms/step - accuracy: 0.3660

- loss: 1.7444 - val_accuracy: 0.5389 - val_loss: 1.2940

Epoch 2/5

1563/1563 ————— **69s** 44ms/step - accuracy: 0.5789

- loss: 1.1934 - val_accuracy: 0.6045 - val_loss: 1.1214

Epoch 3/5

1563/1563 ————— **83s** 45ms/step - accuracy: 0.6373

- loss: 1.0328 - val_accuracy: 0.6542 - val_loss: 1.0045

Epoch 4/5

1563/1563 ————— **80s** 44ms/step - accuracy: 0.6710

- loss: 0.9410 - val_accuracy: 0.6577 - val_loss: 0.9879

Epoch 5/5

1563/1563 ————— **70s** 45ms/step - accuracy: 0.6936

- loss: 0.8796 - val_accuracy: 0.6728 - val_loss: 0.9614

313/313 - 5s - 15ms/step - accuracy: 0.6728 - loss: 0.9614

Test accuracy: 0.67